10.10 (a) From Eq (10.101), the diffracted electric field is given by

$$\vec{E}(\vec{x}) = \frac{1}{2\pi} \nabla \times \int_{\text{apertures}} (\vec{n}' \times \vec{E}) \frac{e^{ikR}}{R} da'$$

Since $R = |\vec{x} - \vec{x}'| = r - \vec{n}\cdot\vec{x}'$, where $\vec{n} = \vec{r}/r$, and when the wave-length is large compared to the dimension of an aperture, $k|\vec{x}'| \ll 1$, we can expand the integral as

$$\vec{E}(\vec{x}) = \frac{1}{2\pi} \nabla \times \int (\vec{n}' \times \vec{E}) \frac{e^{ikr}}{r} (1 - ik\vec{n}\cdot\vec{x}') da'$$

$$= \frac{1}{2\pi} \nabla\left(\frac{e^{ikr}}{r}\right) \times \int (\vec{n}' \times \vec{E})(1 - ik\vec{n}\cdot\vec{x}') da'$$

We are only interested in the radiation field, and only need to retain terms of the order r^{-1}. Then,

$$\vec{E}(\vec{x}) = \frac{ik}{2\pi} \frac{e^{ikr}}{r} \vec{n} \times \int (\vec{n}' \times \vec{E})(1 - ik\vec{n}\cdot\vec{x}') da'$$

$$= \frac{e^{ikr}}{r}\left[\frac{k^2}{2\pi} \vec{n} \times \int (\vec{n}' \times \vec{E})(\vec{n}\cdot\vec{x}') da' + \frac{ik}{2\pi} \vec{n} \times \int (\vec{n}' \times \vec{E}) da' \right]$$

Compare with the expression for scattered field, Eq. (10.2),

$$\vec{E} = \frac{e^{ikr}}{r} \frac{k^2}{4\pi\varepsilon_0} \left[(\vec{n} \times \vec{p}) \times \vec{n} - \frac{1}{c} \vec{n} \times \vec{m} \right],$$

We can immediately recognize the magnetic dipole moment as

$$-\frac{k^2}{4\pi\varepsilon_0} \frac{\vec{m}}{c} = \frac{ik}{2\pi} \int (\vec{n}' \times \vec{E}) da'.$$

or $$\vec{m} = \frac{2\varepsilon_0 c}{ik} \int (\vec{n}' \times \vec{E}) da' = \frac{2\varepsilon_0}{i\omega/c^2} \int (\vec{n}' \times \vec{E}) da' = \frac{2}{i\omega\mu_0} \int (\vec{n}' \times \vec{E}) da'.$$

For the electric dipole, consider the first term in the bracket. Using the result from Section 5.6, it can be shown that

$$\vec{n}\cdot\int \vec{x}'(\vec{n}' \times \vec{E}) da' = -\frac{1}{2} \vec{n} \times \int [\vec{x}' \times (\vec{n}' \times \vec{E})] da' = -\frac{1}{2} \vec{n} \times \int \vec{n}'(\vec{x}'\cdot\vec{E}) da',$$

Since $\vec{x}'\cdot\vec{n}' = 0$. Then,

$$\frac{k^2}{4\pi\varepsilon_0} (\vec{n} \times \vec{p}) \times \vec{n} = -\frac{k^2}{2\pi} \vec{n} \times \left(\frac{1}{2} \vec{n} \times \int \vec{n}'(\vec{x}'\cdot\vec{E}) da' \right).$$

or $$\vec{p} = \varepsilon_0 \vec{n}' \int (\vec{x}'\cdot\vec{E}) da',$$ as $\vec{n}'$ is a constant vector for the entire aperture.

(b) In the vector identity $\nabla(\psi\vec{a}) = (\vec{a}\cdot\nabla)\psi + \psi\,\nabla\cdot\vec{a}$, replace ψ with a vector and let $\psi = \vec{x}'$, and $\vec{a} = \vec{n}'\times\vec{E}$, then we know

$$\nabla\left[\vec{x}'(\vec{n}'\times\vec{E})\right] = \left[(\vec{n}'\times\vec{E})\cdot\nabla\right]\vec{x}' + \vec{x}'\,\nabla\cdot(\vec{n}'\times\vec{E}) = \vec{n}'\times\vec{E} + \vec{x}'\,\nabla\cdot(\vec{n}'\times\vec{E})$$

Therefore, $$\int(\vec{n}'\times\vec{E})\,da' = \int\nabla\left[\vec{x}'(\vec{n}'\times\vec{E})\right]da' - \int\vec{x}'\left[\nabla\cdot(\vec{n}'\times\vec{E})\right]da'$$

The first term is a total differential and reduces to an integral around the aperture, which can be dropped as the incident field is zero at the edge. In the second term,

$$\nabla\cdot(\vec{n}'\times\vec{E}) = \vec{E}\cdot(\nabla\times\vec{n}') - \vec{n}'\cdot(\nabla\times\vec{E}) = -\vec{n}'\cdot(\nabla\times\vec{E}) = \vec{n}'\cdot\frac{\partial\vec{B}}{\partial t} = -i\omega(\vec{n}'\cdot\vec{B}),$$

where we have used the fact that $\vec{n}'$ is a constant. Finally,

$$\vec{m} = \frac{2}{i\omega\mu_0}\int(\vec{n}'\times\vec{E})\,da' = -\frac{2}{i\omega\mu_0}\int\vec{x}'\left[\nabla\cdot(\vec{n}'\times\vec{E})\right]da' = \frac{2}{\mu_0}\int\vec{x}'(\vec{n}'\cdot\vec{B})\,da'.$$